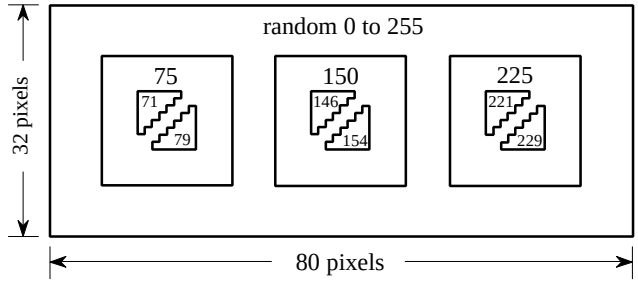

FIGURE 23-11
Brightness and contrast test image. This is the structure of the digital image used in Fig. 23-12. The three squares form dark, medium, and bright objects, each containing two low contrast triangles. This figure indicates the digital numbers (DN) of the pixels in each region.

words, there is a dark region in the image with faint detail, this is a medium region in the image with faint detail, and there is a bright region in the image with faint detail.

Figure 23-12 shows how adjustment of the contrast and brightness allows different features in the image to be visualized. In (a), the brightness and contrast are set at the *normal* level, as indicated by the **B and C slide bars** at the left side of the image. Now turn your attention to the graph shown with each image, called an **output transform**, an **output look-up table**, or a **gamma curve**. This controls the hardware that displays the image. The value of each pixel in the stored image, a number between 0 and 255, is passed through this look-up table to produces another number between 0 and 255. This new digital number drives the video intensity circuit, with 0 through 255 being transformed into black through white, respectively. That is, the look-up table maps the stored numbers into the displayed brightness.

Figure (a) shows how the image appears when the output transform is set to do *nothing*, i.e., the digital output is identical to the digital input. Each pixel in the noisy background is a random shade of gray, equally distributed between black and white. The three boxes are displayed as dark, medium and light, clearly distinct from each other. The problem is, the triangles inside each square cannot be easily seen; the contrast is too low for the eye to distinguished these regions from their surroundings.

Figures (b) & (c) shows the effect of changing the brightness. Increasing the brightness shifts the output transform to the *left*, while decreasing the brightness shifts it to the *right*. Increasing the brightness makes *every* pixel in the image appear lighter. Conversely, decreasing the brightness makes *every* pixel in the image appear darker. These changes can improve the viewability of excessively dark or light areas in the image, but will **saturate** the image if taken too far. For example, all of the pixels in the far right square in (b) are displayed with full intensity, i.e., 255. The opposite effect is shown in (c), where all of the pixels in the far left square are displayed as blackest black, or digital number 0. Since all the pixels in these regions have the same value, the triangles are completely wiped out. Also notice that *none* of the triangles in (b) and (c) are easier to see than in (a). Changing the brightness provides little (if any) help in distinguishing low contrast objects from their surroundings.